ALLEN

(Academic Session: 2026 – 2027)

TALLENTEX PRACTICE QUESTIONS

CLASS 10

MATHEMATICS

Your Hard Work Leads to Strong Foundation

Practice Questions

1. If n is a natural number, then 12^n will always end with an even digit except

(1) 4 (2) 6 (3) 8 (4) 0

2. Jack and Jill exercise along the same route. Jill jogs the first half of the route at 6 km/h, runs the remaining route at 12 km/h and takes a total time of x hours. Jack walks the first third of the route at 5 km/h, runs the remaining at 15 km/h and takes a total time of y hours. Which of the following is equal to $\frac{x}{y}$?

(1) $\frac{9}{8}$ (2) $\frac{7}{5}$ (3) $\frac{15}{16}$ (4) $\frac{9}{16}$

3. The points (0, –2), (3, 0) and (–3, –4) lie on a graph of a linear polynomial, then the zero of the polynomial is

(1) 0 (2) –2 (3) 3 (4) –3

4. How many planes can be made to pass through three distinct points?

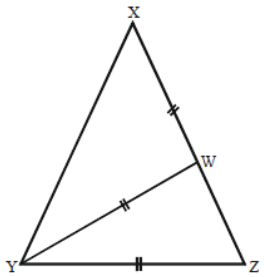
(1) one plane

(2) two plane

(3) no plane

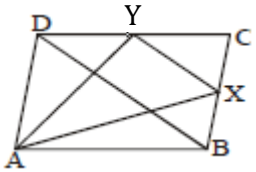
(4) infinite if they are collinear and only one if they are non collinear.

5. In the diagram, ΔXYZ is isosceles with XY = XZ. Also, point W is on XZ so that XW = WY = YZ. The measure of $\angle XYW$ is



(1) 18° (2) 36° (3) 45° (4) 30°

6. ABCD is a parallelogram X and Y are the mid-points of BC and CD respectively. Then the area of ΔAXY is equals to



(1) $\frac{1}{8}$ ar(parallelogram ABCD) (2) $\frac{2}{8}$ ar(parallelogram ABCD)

(3) $\frac{3}{8}$ ar(parallelogram ABCD) (4) $\frac{4}{8}$ ar(parallelogram ABCD)

7. $(\cos 0° + \sin 30° + \sin 45°)(\sin 90° + \cos 60° - \cos 45°) = ?$

(1) $\frac{5}{6}$ (2) $\frac{5}{8}$ (3) $\frac{3}{5}$ (4) $\frac{7}{4}$

8. The sides of a triangle are in the ratio 12 : 17 : 25 and its perimeter 540 cm. What is its area ?

(1) 1.4 m^2 (2) 0.9 m^2 (3) 0.7 m^2 (4) 1.2 m^2

9. If the mean of the squares of first n natural numbers is 105, then find the median of the first n natural numbers.

(1) 8 (2) 9 (3) 10 (4) 11

10. $2(\sin^6 q + \cos^6 q) - 3(\sin^4 q + \cos^4 q) + 1 =$

(1) 1 (2) 0 (3) –1 (4) –2

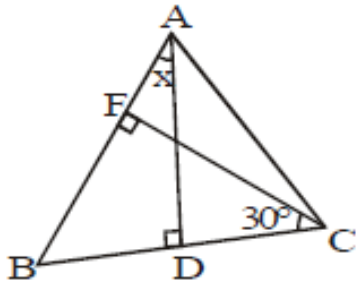
11. If the distance between the points (4, p) and (1, 0) is 5 units, then p =

(1) ±4 (2) 4 (3) –4 (4) 0

12. One equation of a pair of inconsistent linear equations is x – 3y – 5 = 0, then the second equation can be

(1) 2x – 6y = 10 (2) –2x +6y = 5 (3) 3x +9y = 5 (4) x + 3y = 5

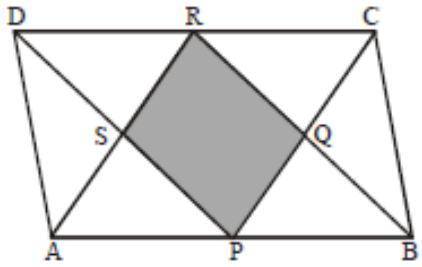
13. Consider the following figure



What is the measure of angle x?

(1) 20° (2) 30° (3) 40° (4) 50°

14. In the adjoining figure ABCD, P and R are the mid-points of the sides AB and CD. ABCD is a parallelogram. What is the ratio of the shaded to the unshaded region ?



(1) 1 : 2 (2) 1 : 3 (3) 1 : 4 (4) none of these

15. If tanx = 3cot x then x = ?

(1) 45° (2) 60° (3) 30° (4) 15°

16. The base of an isosceles right triangle is 30 cm. Its area is

(1) 225 sq. cm (2) $225\sqrt{3}$ sq. cm (3) $225\sqrt{2}$ sq. cm (4) 450 sq. cm

17. If the weight of a spherical shell is $\frac{7}{8}^{th}$ of what it would be if it were a solid sphere. The ratio of inner to outer radii of the shell is

(1) 1 : 2 (2) 1 : 3 (3) 2 : 3 (4) 3 : 4

18. The mean of a set of 20 observations is 8 and of another set of 30 observations is 10. The mean of combined set is

(1) 9.2 (2) 10.8 (3) 11.2 (4) 9.8

19. The value of [(secA + tanA)(1 – sinA)] is equal to

(1) $\tan^2 A$ (2) $\sin^2 A$ (3) cosA (4) sinA

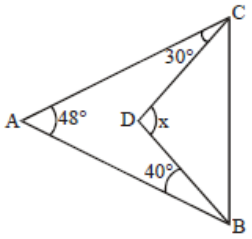
20. How many positive integers less than 100 have reciprocals with terminating decimal representations?

(1) 13 (2) 14 (3) 21 (4) 25

21. Ram, Shyam, Tarun and Varun together had a total amount of Rs 240 with them. Ram had half of the total amount with the others. Shyam had one-third of the total amount with the others. Tarun had one-fourth of the total amount with the others. Find the amount with Varun (in Rs).

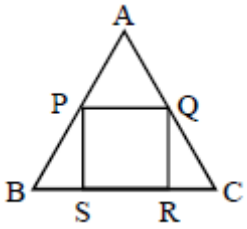
(1) 64 (2) 70 (3) 52 (4) 58

22. Find the value of x in the figure given below.



(1) 118° (2) 20° (3) 72° (4) 223°

23. In the figure given below, ABC is an equilateral triangle and PQRS is a square of side 6 cm. By how many cm^2 is the area of the triangle more than that of the square?



(1) $\frac{21}{\sqrt{3}}$ (2) 21 (3) $21\sqrt{3}$ (4) 63

24. Evaluate : $\cot^2 30° - 2\cos^2 30° - \frac{3}{4}\sec^2 45° + \frac{1}{4}\text{cosec}^2 30°$

(1) 1 (2) 0 (3) –1 (4) 2

25. In a triangle if each side of a triangle is halved, then what is the percentage change in its area.

(1) 75% increase (2) 75% decrease (3) 25% increase (4) 25% decrease

26. The mean of 90 items was found to be 45. Later on, it was discovered that two items were misread as 26 and 19 instead of 62 and 09 respectively. The correct mean is (approximately)

(1) 49.0 (2) 45.0 (3) 45.3 (4) 49.3

27. If the points A(4, 3) and B(x, 5) are on the circle with centre O(2, 3),find the value of x.

(1) 1 (2) 2 (3) 3 (4) 4

28. The product of 2 whole numbers is 1000. If neither of the number is a multiple of 10. What is their sum ?

(1) 110 (2) 125 (3) 150 (4) 133

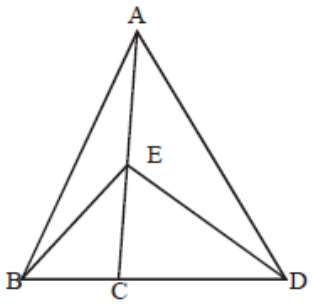
29. I have some tables and chairs. If I place two chairs at each table, I have one extra chair. If I place three chairs at each table, I have one table with no chairs. What is the sum of the total number of tables and chairs?

(1) 12 (2) 7 (3) 17 (4) 13

30. One of the exterior angles of an isosceles triangle is 150°. What is the ratio of its unequal interior angles in same triangle ?

(1) 1 : 4 (2) 5 : 2 (3) either (1) or (2) (4) Can't say

31. If BC : CD = 2 : 3, AE : EC = 3 : 4 and BC : AE = 2 : 3, then find the ratio of the area of ΔECD to the area of ΔAEB.



(1) 2 : 1 (2) 2 : 3 (3) 3 : 5 (4) 4 : 3

32. Find the ratio in which the point (–3, p) divides the line segment joining the points (–5, –4) and (–2, 3).

(1) 2 : 1 (2) 1 : 2 (3) 2 : 3 (4) 3 : 2

33. The perimeter of a right angled triangle is 450 m. If its sides are in the ratio 5 : 12 : 13, then the area of the triangle is

(1) 9000 m^2 (2) 8765 m^2 (3) 6750 m^2 (4) 11750 m^2

34. In a class of 15 students, on an average, each student got 12 books. If exactly two students received same number of books, and remaining student books average be an integer then which of the following could be the number of books received by each of the two students who received same number of books?

(1) 11 (2) 15 (3) 20 (4) 25

35. Find the coordinates of the centre of the circle passing through the points (0, 0), (–2, 1) and (–3, 2).

(1) (3, 11) (2) $\left(\frac{3}{2},\frac{11}{2}\right)$ (3) $\left(\frac{3}{2},\frac{5}{2}\right)$ (4) (0, 5)

36 One of the factor of $(a + 2b)^3 + (2a – c)^3 – (a + 2c)^3 + 3(a + 2b)(2a – c)(a + 2c)$ is

(1) 2a + 2b - 3c (2) 2a – 2b + 3c (3) 2a + 2b + 3c (4) –2a – 2b – 3c

37. Which of the following sets of numbers are relative primes

(i) 51, 85 (ii) 26, 65 (iii) 57, 76 (iv) 29, 75

(1) only (ii) (2) only (iv) (3) both (i) and (iii) (4) All of these

38. In a family of husband, wife and a daughter, the sum of the husband's age, twice the wife's age and thrice the daughters age is 85; while the sum of twice the husband's age, 4 times the wife's age and 6 times the daughter's age is 170. It is also given that the sum of 5 times the husband's age, ten times the wife's age and 15 times the daughter's age equals 450. The number of possible solutions, in terms of the ages of the husband, wife and the daughter, to this problem is

(1) 0 (2) l (3) 2 (4) infinitely many

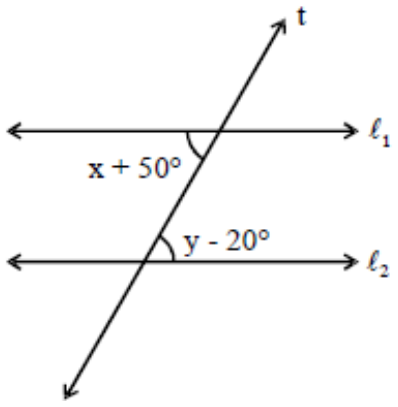
39. If (2a+b) = 12 and ab = 15, then the value of $8a^3 + b^3$ is

(1) 640 (2) 1080 (3) 616 (4) 648

40. 'Lines are parallel if they do not intersect' is stated in the form of

(1) an axiom (2) a definition (3) a postulate (4) a proof

41. In the given figure, $\ell_1 \parallel \ell_2$ and x and y are complementary angles. Find x & y



(1) 50°, 40° (2) 20°, 70° (3) 10°, 80° (4) 30°, 60°

42. In ΔABC, AD is median and P is a point in AD such that AP : PD = 1 : 2 then area of ΔABP =

(1) $\frac{1}{2}$ ar (ΔABC) (2) $\frac{2}{3}$ ar (ΔABC) (3) $\frac{1}{3}$ ar (ΔABC) (4) $\frac{1}{6}$ ar (ΔABC)

43. The three vertices of a parallelogram taken in order are (–1, 0), (3,1) and (2, 2) respectively. Find the coordinates of fourth vertex.

(1) (2, 1) (2) (1, 2) (3) (–2, 1) (4) (1, –2)

44. In a triangle, the sides are given as 11 cm, 12 cm and 13 cm. The approximate length of the altitude corresponding to the side having length 12 cm.

(1) 10.5 cm (2) 10.25 cm (3) 10 cm (4) 7.25 cm

45. A cone, a hemisphere and a cylinder stand on equal bases of radius R and have equal heights H. Their total surface area are in the ratio

(1) $\left(\sqrt{3}+1\right):3:4$ (2) $\left(\sqrt{2}+1\right):7:8$ (3) $\left(\sqrt{2}+1\right):3:4$ (4) None of these

46. Find the value of '2x' if the Mean of x, x + 2, x + 7, x + 11 and 2x is 16

(1) 10 (2) 20 (3) 30 (4) 40

47. The G.C.D. of two polynomials is (x – 1) and their L.C.M is $x^6 - 1$. If one of the polynomials is $x^3 - 1$, then the other polynomial is .

(1) $x^3 - 1$ (2) $x^4 - x^3 + x - 1$ (3) $x^2 - x + 1$ (4) None of these

48. Value of x, y which satisfies 3x + 5y = 12xy and 7x – 2y = 4xy are

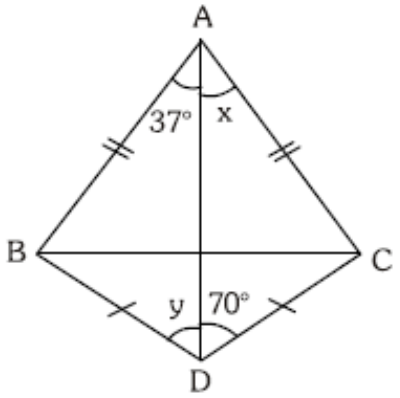
(1) x = $\frac{37}{31}$, y = $\frac{41}{31}$ (2) y = $\frac{41}{44}$, x = $\frac{41}{72}$

(3) x = 1, y = 2 (4) y = $\frac{32}{41}$, x = $\frac{44}{41}$

49. If we draw the graph of a cubic polynomial, then it will intersect the axis of x at least in

(1) Zero point (2) One point (3) Two points (4) Three points

50. In the given figure, what are the values of x and y?



(1) x = 70°, y = 37° (2) x = 37°, y = 70° (3) x + y = 117° (4) x – y = 100°

51. In the parallelogram ABCD, the side AB is produced to the point X, so that BX = AB. The line DX cuts BC at E. Area of ΔAED =

(1) 2× area (ΔCEX) (2) $\frac{1}{2}$ × area (ΔCEX) (3) area (ΔCEX) (4) $\frac{1}{3}$ × area (ΔCEX)

52. The length of three sides of a triangular field is 30 m, 16 m and 18 m respectively. The cost of fencing per meter is Rs. 5. Then find total cost of fencing around the field

(1) Rs. 320 (2) Rs. 300 (3) Rs. 325 (4) Rs. 400

53. The average weight of 5 men is decreased by 3 kg when one of them weighing 150 kg is replaced by another person. This new person is again replaced by another person whose weight is 30 kg lower than the person he replaced. What is the overall change in the average due to this dual change ?

(1) 6 kg (2) 9 kg (3) 12 kg (4) 15 kg

54. An octopus has 8 tentacles and 1 head. A jellyfish has 20 tentacles and no head. A cow has 4 legs and 1 head. Farmer Brown has a total of 17 heads, 196 tentacles and 20 legs. So, how many animals does she have

(1) 20 (2) 22 (3) 30 (4) None of these

55. A deck of 100 cards is numbered from 1 to 100. Each card has the same number printed on both sides. One side of each card is red and the other side is yellow. Bobby places all the cards, red side up, on a table. He first turns over every card that has a number divisible by 2. He then examines all the cards, and turns over every card that has a number divisible by 3. How many cards have the red side up when Bobby is finished?

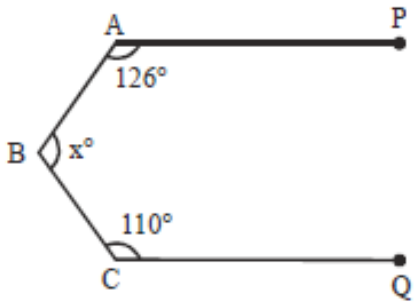
(1) 83 (2) 49 (3) 66 (4) 50

56. The three steps from solid to point are

(1) Solid - surface - line - point (2) Line - point - surface - solid

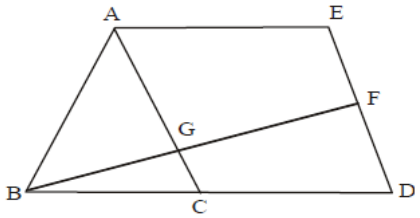
(3) Surface - point - line - solid (4) Point - surface - line - solid

57. Find x in the following figure



(1) 110° (2) 86° (3) 124° (4) 135°

58. In the given figure (not to scale), ABC is an isosceles triangle in which AB = AC. AEDC is a parallelogram. If $\angle CDF = 70°$ and $\angle BFE = 100°$, then find $\angle FBA$.



(1) 30° (2) 40° (3) 50° (4) 80°

59. An equilateral triangle has side $2\sqrt{3}$ cm. The radius of its circumcircle will be :

(1) 2 cm (2) $\sqrt{3}$ cm (3) 3 cm (4) 4 cm

60. In a garden, a triangular piece of land has to be planted with grass which cost Rs. 5 per 100 cm^2. The sides of the triangle are 8m, 6m and 6m. Find the total cost of planting the grass. Take $\left(\sqrt{5} = 2.23\right)$

(1) Rs. 8920 (2) Rs. 7500 (3) Rs. 8000 (4) None of these

61. The length of the longest rod that can be placed in a room which is 12m long, 9m broad and 8m high is

(1) 27m (2) 19m (3) 17m (4) 13m

62. Simplify : $\left(x+\frac{1}{5}\right)(x+5)$

(1) $x^2 + 5.2x + 1$ (2) $x^2 + x + 1$ (3) $x^2 + 5.2x + 25$ (4) $x^2 + 1$

63. Four of the six numbers 1867, 1993, 2019, 2025, 2109 and 2121 have a mean of 2008. What is the mean of the other two numbers.

(1) 1994 (2) 2006 (3) 2022 (4) 2051

64. A boat can travel with a speed of 13 km/hr in still water. If the speed of the stream is 4 km/hr, find the time taken by the boat to go 68 km downstream.

(1) 2 hours (2) 3 hours (3) 4 hours (4) 5 hours

65. In a ΔABC, P, Q, R are mid points of sides BC, CA and AB respectively. If AC = 16 cm, BC = 20 cm and AB = 20 cm, then the perimeter of $\square ARPQ$ will be :

(1) 60 cm (2) 30 cm (3) 40 cm (4) None of these

66. In ΔABC if D is a point in BC and divides it in the ratio 3 : 5 then ar (ΔADC) : ar(ΔABC) =

(1) 3 : 5 (2) 3 : 8 (3) 5 : 8 (4) 8 : 3

67. The perimeter of a triangle is 50 cm. One side of a triangle is 4 cm longer than the smaller side and the third side is 6 cm less than twice the smaller side. Find the area of the triangle?

(1) $20\sqrt{30}$ cm^2 (2) 20 cm^2 (3) 30 cm^2 (4) $35\sqrt{30}$ cm^2

68. The areas of three adjacent faces of a cuboid are $1m^2$, $4m^2$ and $9m^2$ then its volume is

(1) 36 m^3 (2) 6 m^3 (3) 216 m^3 (4) Can't be determined

69. If $2a+\frac{1}{3a}=6$, then the value of the expression $3a+\frac{1}{2a}$ is,

(1) 12 (2) 9 (3) 4 (4) 8

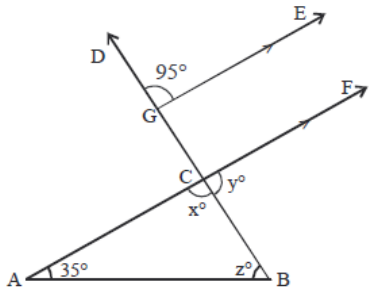
70. The mean of a set of observation is a. If each observation is multiplied by b and each product is decreased by c, then the mean of new set of observations is_____.

(1) $\frac{a}{b}+c$ (2) ab – c (3) $\frac{a}{b}-c$ (4) ab + c

71. From 87x + 53y = 15; 53x + 87y = – 155, find the value of x – y is _____

(1) 4 (2) 5 (3) 3 (4) can't be determined

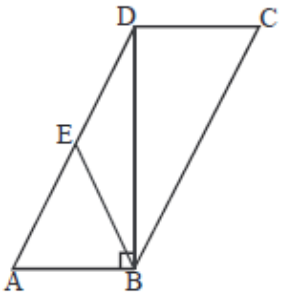
72. From the given figure, find x, y, z if GE || AF.



(1) 95°, 58°, 50° (2) 85°, 58°, 50° (3) 95°, 85°, 95° (4) 95°, 85°, 50°

73. In the given figure, □ABCD is a parallelogram, E is midpoint of AD.

$\frac{ar(\Delta BED)}{ar(||^{gm}\ ABCD)}=$ _____



(1) $\frac{4}{3}$ (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

74. A floral design of a floor is made up of 16 tiles which are triangular in shape. The side of the triangle being 9 cm, 28 cm and 35 cm. The cost of polishing the tiles at 50 paisa/sq. cm is _____$\left(\sqrt{6}=2.45\right)$

(1) Rs. 289 (2) Rs. 288 (3) Rs. 705.6 (4) Rs. 500

75. The radius of two spheres is in the ratio 4 : 5 then, the ratio of their volumes is
(1) 4:5 (2) 64:125 (3) 16:25 (4) 5:4

76. Factorize the following polynomial $x + 8xy^3$.
(1) $x(1 - 2y)(1 - 2y + 4y^2)$ (2) $x(1 + 2y)(1 - 2y + 4y^2)$
(3) $x(1 + 2y)(1 + 2y + 4y^2)$ (4) $x(1 - 2y)(1 + 2y + 4y^2)$

77. Mean weight of a group of 30 students is 62.5 kg. A student (weight 71.5 kg) left the group and a new student (weight 53.5 kg) joined the group. The new mean of the group will be
(1) 60 kg (2) 63.5 kg (3) 61.9 kg (4) 59.5 kg

78. The real numbers x and y are such that $x+\frac{2}{y}=\frac{8}{3}$ and $y+\frac{2}{x}=3$. The value of xy is
(1) $\frac{4}{3}$ (2) $\frac{16}{9}$ (3) 2 (4) 4

79. How many three digit numbers are there in which all the digits are odd?
(1) 100 (2) 125 (3) 500 (4) 250

80. In a right triangle ACB with sides a and b, and hypotenuse c, the altitude drawn to the hypotenuse from C is x. Then which one of the following is correct ?
(1) $ab = x^2$ (2) $\frac{1}{a}+\frac{1}{b}=\frac{1}{x}$ (3) $a^2 + b^2 = 2x^2$ (4) $\frac{1}{x^2}+\frac{1}{a^2}=\frac{1}{b^2}$

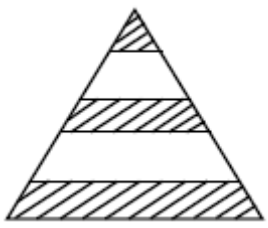
81. In a parallelogram ABCD, M and N are the mid-points of sides AD and BC respectively. If MN = 10 cm and length of perpendicular from point 'A' to CD is 6 cm, then the area of the parallelogram ABCD is
(1) 30 cm^2 (2) 60 cm^2 (3) 90 cm^2 (4) None of these

82. If the length of the sides of a triangle are 8 cm, 11 cm and 13 cm, then the area of the triangle is (in cm^2) :
(1) 44 (2) 43 (3) 42.82 (4) $8\sqrt{30}$

83. The radius of a sphere is 2r, then its volume will be
(1) $\frac{4}{3}\pi r^3$ (2) $\frac{8}{3}\pi r^3$ (3) $4\pi r^3$ (4) $\frac{32}{3}\pi r^3$

84. If the mode of 4, 9, 5, 4, 9, 5, 4, 9 and x – 10 is 9, then the value of x is
(1) 4 (2) 19 (3) 1 (4) 5

85. In adjoining figure (Not to scale), the triangle is divided into five parallel strips of equal height. The total area of the shaded stripes is 45 square unit. What is the total area of the unshaded strips?



(1) 30 sq. unit (2) 60 sq. unit (3) 45 sq. unit (4) 15 sq. unit

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**1**	**3**	**4**	**2**	**3**	**4**	**2**	**2**	**2**	**1**	**2**	**2**	**2**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**1**	**1**	**3**	**2**	**3**	**1**	**3**	**1**	**2**	**3**	**2**	**4**	**4**	**3**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**1**	**1**	**3**	**4**	**2**	**1**	**2**	**1**	**4**	**2**	**3**	**4**	3	**2**	**3**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**2**	**2**	**2**	**2**	**2**	**1**	**1**	**1,2**	**2**	**2**	**1**	**3**	**2**	**1**	**1**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**3**	**1**	**4**	**3**	**4**	**3**	**1**	**2**	**2**	**2**	**2**	**4**	**2**	**3**	**2**
Question	76	77	78	79	80	81	82	83	84	85					
Answer	**2**	**3**	**3**	**2**	**4**	**2**	**4**	**4**	**2**	**1**					